Sunmook Choi, 최선묵

https://smfelixchoi.github.io sc3377@cornell.edu

Research Interests

I study bandits, reinforcement learning, optimal control using tools and concepts from control theory, statistical learning theory, and optimization.

Education

Cornell University

- PhD Candidate, Center for Applied Mathematics — Aug. 2024 – Present
  PhD Minors: Mathematics, Computer Science
  Thesis Committee: Sarah Dean (Advisor), David Bindel (Math), Robert Kleinberg (CS)

Korea University

- M.S., Mathematics — Mar. 2022 – Aug. 2024
  Thesis: Three-dimensional dicube tilings of augmented Aztec bipyramids
  Advised by Prof. Seungsang Oh
- B.S., Mathematics — Mar. 2016 – Feb. 2022
- Korean Mandatory Military Service — Jun. 2017 – Feb. 2019

Honors & Awards

- KUAA 2024 Scholarship (Korea University Alumni Association in NY) — Dec 2024
- Editors' Choice Award 2023, Discrete Mathematics — Mar 2024
- IJCAI 2023 Competition and Challenge Award — Aug 2023
- ICASSP 2022 Grand Challenge Award — Mar 2022
- Sigma Scholarship (Korea Univ. Dept. of Mathematics Alumni Association) — Fall 2021
- Dean's List, College of Science, Korea University — Fall 2020
- Semester High Honors, Korea University — Spring (2016,2019,2020,2021), Fall (2019,2020,2021)
- SAMSUNG Welfare Foundation Scholarship (Dream Class) — Spring 2019

Selected Publications

A full list of papers can be found in Google Scholar.

Preprint

[1] *Performance Gaps Between Open-Loop, Certainty-Equivalent, and Optimal Control Under Action-Dependent Observations.*
Leo Maynard-Zhang, **Sunmook Choi**, Yahya Sattar, Yassir Jedra, Sarah Dean, Maryam Fazel

[2] *Dual Control of Linear Systems from Bilinear Observations with Belief Space Model Predictive Control.*
arXiv:2604.24663
Daniel Cao, Beixi Du, Andrew Lowitt, **Sunmook Choi**, Sarah Dean, Yahya Sattar

Conference Papers

[1] *Two-Layer Linear Auto-Regressive Models Estimate Latent States*
International Conference on Machine Learning (ICML), 2026
Yahya Sattar, **Sunmook Choi**, Leo Maynard-Zhang, Yassir Jedra, Maryam Fazel, Sarah Dean

[2] *Explore-then-Commit for Nonstationary Linear Bandits with Latent Dynamics*
International Conference on Artificial Intelligence and Statistics (AISTATS), 2026
Sunmook Choi, Yahya Sattar, Yassir Jedra, Maryam Fazel, Sarah Dean

[3] *Sub-optimality of the Separation Principle for Quadratic Control from Bilinear Observations*
IEEE Conference on Decision and Control (CDC), 2025
Yahya Sattar, **Sunmook Choi**, Yassir Jedra, Maryam Fazel, Sarah Dean

[4] *TB-ResNet: Bridging the gap from TDNN to ResNet in Speaker Verification with Temporal-Bottleneck Enhancement*
IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2024
Sunmook Choi, Sanghyeok Chung, Seungeun Lee, Soyul Han, Taein Kang, Jaejin Seo, Il-Youp Kwak, Seungsang Oh

[5] *Overlapped Frequency-Distributed Network: Frequency-Aware Voice Spoofing Countermeasure*
Interspeech, 2022
Sunmook Choi, Il-Youp Kwak, Seungsang Oh

[6] *Light-weight Frequency Information Aware Neural Network Architecture for Voice Spoofing Detection*
International Conference on Pattern Recognition (ICPR), 2022
Sunmook Choi, Seungsang Oh, Jonghoon Yang, Yerin Lee, Il-Youp Kwak

Workshop Papers

[1] *CAU KU deep fake detection system for ADD 2023 challenge*
Workshop on Deepfake Audio Detection and Analysis at IJCAI, 2023
Soyul Han*, Taein Kang*, **Sunmook Choi***, Jaejin Seo, Sanghyeok Chung, Seungeun Lee, Seungsang Oh, Il-Youp Kwak
(*: equal contribution)

[2] *Low-quality Fake Audio Detection through Frequency Feature Masking*
International Workshop on Deepfake Detection for Audio Multimedia at ACM Multimedia, 2022
Il-Youp Kwak, **Sunmook Choi**, Jonghoon Yang, Yerin Lee, Soyul Han, Seungsang Oh

Journal Articles

[1] *Augmented Aztec bipyramid and dicube tilings*
Sunmook Choi, Sangyop Lee, Seungsang Oh
Discrete Mathematics, 2024, **Editor's Choice**

Thesis

[1] *Three-dimensional dicube tilings of augmented Aztec bipyramids*
Master's thesis, Korea University DOI link

Teaching

Teaching Assistant, Dept. of Mathematics, Cornell University

- Linear Algebra (MATH 4310) — Spring 2026
- Linear Algebra for Engineers (MATH 2940) — Fall 2025
- Linear Algebra for Engineers (MATH 2940) — Spring 2025
- Linear Algebra for Engineers (MATH 2940) — Fall 2024

Teaching Assistant, Dept. of Mathematics, Korea University

- Linear Algebra II With LAB (MATH 222) — Fall 2023
- Linear Algebra I With LAB (MATH 221) — Spring 2023
- Linear Algebra II With LAB (MATH 222) — Fall 2022
- Linear Algebra I With LAB (MATH 221) — Spring 2022

Service & Leadership

Reviewer: AISTATS, NeurIPS, TMLR, ICML

- Recognized as a *Top Reviewer* from NeurIPS 2026.
- Designated as a *Gold Reviewer* (top 25%) from ICML 2026.

Research Mentor

- Undergraduate Independent Research, Cornell — Fall 2026
- The Bowers Undergraduate Research Experience (BURE) Program — Summer 2026
- M.Eng. Program, Cornell — Fall 2025 – Spring 2026

Reading Mentor, Directed Reading Program, Math Department, Cornell — Spring 2026

- Guided an undergraduate student through mathematical reading and weekly discussions

Program Mentor, CURIE Academy, Cornell Engineering High School Outreach Program — Summer 2025

- Control & Navigation of Weather Balloons, led by Prof. Sarah Dean

Seminar Leader, Deep Learning Seminar, Season 1 & 2, DLmath — Jun. 2023 – Aug. 2023, Sep. 2022 – Feb. 2023

- Theoretical Review and Python Implementation of SGD, MLP, CNN, RNN, AE, VAE, GAN.
- Season 1 materials, Season 2 materials

Seminar Leader, Deep Reinforcement Learning Seminar, — Mar. 2023 – Sep. 2023

- Theoretical Review and Python Implementation of MDP, DP, RL, DQN, PG, AC, PPO.
- Materials

Tutor, SAMSUNG Dream Class — Spring 2019

- Teaching Mathematics to middle school students as an after-school program.

Competences

Languages Korean, English

Techniques LaTeX, Python, TensorFlow, PyTorch, R, Julia